Wheel Odometry and Error Tracking

Assignment 1: Localization and Simulation Foundations | Keran Chen

Ubuntu 24.04 | ROS 2 Jazzy | Gazebo Sim 8.15.0

Problem Statement

Track the planar pose (x, y, heading) of a differential-drive robot in an indoor simulated room. The robot executes four nominal 2 m straight sections and four 90-degree turns. The aim is to measure accumulated wheel-odometry error and isolate the effect of added encoder noise.

Methods

The estimator uses measured wheel-joint angles only, starting from the known configured pose (0, 0, 0). Angles are quantised to 2,048 counts/revolution and converted to wheel travel using radius r = 0.15 m; wheel separation b = 0.625 m. The Lecture 2 short-step midpoint update runs at approximately 50 Hz:

```
ds = (dsR + dsL)/2; dtheta = (dsR - dsL)/b
x_next = x + ds*cos(theta + dtheta/2)
y_next = y + ds*sin(theta + dtheta/2)
theta_next = wrap(theta + dtheta)
```

Wheel measurements follow d_measured = (1 + bias)d + epsilon, with independent Gaussian epsilon of standard deviation k sqrt(|d|). Left/right scale biases are 0%/1%. These are declared experimental assumptions; k has units m/sqrt(m). The same saved encoder trajectory is replayed twice: baseline k = 0 and noisy k = 0.01, with all other settings fixed and seed 42.

Simulator pose is used only for evaluation. Both estimates are generated before ground truth is read. Truth is interpolated to estimate timestamps in the same world frame; heading interpolation and differences use the shortest angle. Evaluation starts at first encoder motion and ends one second after last motion, excluding the extended idle tail.

Results

Metric	Baseline (k = 0)	Noisy (k = 0.01)
Final position error (m)	0.2218	0.1808
Position RMSE (m)	0.1459	0.1073
Final signed heading error (deg)	9.150	8.886

Over 52.84 s and 2,631 matched samples per condition, added noise decreased final position error by 0.0410 m for this realisation. The baseline already drifts because of the fixed wheel-scale mismatch and wheel/ground motion differences. Heading errors rotate subsequent displacement estimates, producing growing trajectory separation. Random noise need not worsen every endpoint; one route and one seed do not establish a statistical trend.

Evidence: results/trajectory_comparison.png (labelled truth and estimate trajectories), results/error_comparison.png, and results/metrics.json. Reproduce with python3 scripts/compare_noise.py; compressed encoder and ground-truth inputs are included. Local Git history records simulation, estimation and evaluation development.